RDBMS Interview Questions

1. What is an RDBMS?
2. Explain the key features of an RDBMS.
3. What is a relational database?
4. Differentiate between a database and an RDBMS.
5. What are the advantages of using an RDBMS?
6. What is a primary key? How is it different from a unique key?
7. Explain the concept of normalization and its importance in RDBMS.
8. What are the different types of relationships in a relational database?
9. Define the terms "foreign key" and "referential integrity."
10. What is a query in an RDBMS? Provide an example.
11. What is ACID in the context of RDBMS?
12. What are the different types of joins in SQL?
13. How do you optimize SQL queries for performance?
14. Explain the concepts of indexes and their types.
15. What is a transaction? What are the properties of a transaction in RDBMS?
16. How does an RDBMS handle concurrency and ensure data integrity?
17. Describe the concept of data normalization and its different normal forms.
18. What is a view in an RDBMS? How is it different from a table?
19. What are stored procedures and triggers? How are they used in an RDBMS?
20. Discuss the role of the RDBMS in data security and access control.
21. What is the difference between a database and a file oriented system?
22. Explain the concept of data integrity and its significance in an RDBMS.
23. What is the purpose of a foreign key constraint in a relational database?
24. Describe the concept of database normalization and its benefits.
25. How do you enforce referential integrity between tables in an RDBMS?
26. What is a deadlock in the context of database management systems? How can it be prevented?
27. Explain the differences between DELETE, TRUNCATE, and DROP statements in SQL.
28. What are aggregate functions in SQL? Provide examples of some commonly used aggregate functions.

29. What is the purpose of the GROUP BY clause in SQL? How is it used?
30. Discuss the differences between a clustered and non-clustered index in an RDBMS.
31. What is the purpose of a transaction log in an RDBMS? How does it ensure data consistency?
32. How can you improve the performance of a database query with proper indexing?
33. Explain the concept of database normalization and its different normal forms.
34. What is a stored procedure? What are its advantages in an RDBMS?
35. How does an RDBMS handle concurrent access to data? Explain the concept of locking and its types.
36. What is the role of the COMMIT and ROLLBACK statements in an RDBMS?
37. Discuss the differences between a heap and a clustered table in an RDBMS.
38. What is the purpose of the HAVING clause in SQL? How is it different from the WHERE clause?
39. Explain the concept of database replication and its benefits.
40. How does an RDBMS handle data backup and recovery?
41. Explain the ACID properties in the context of database transactions and discuss their significance.
42. What is the difference between a candidate key, a primary key, and a superkey in an RDBMS?
43. Describe the concept of database indexing. What are the different types of indexes and their trade-offs?
44. Discuss the various types of database joins in SQL, including inner join, left join, right join, and full outer join.
45. Explain the concept of normalization and its different normal forms (1NF, 2NF, 3NF, BCNF, etc.). Provide examples.
46. Difference between 3NF & BCNF.
47. What are triggers in an RDBMS? How do they work, and when are they used?
48. Describe the concept of database replication. What are the different replication methods and their purposes?
49. Discuss the advantages and disadvantages of using stored procedures in an RDBMS.
50. Explain the concept of database denormalization. When and why would you denormalize a database?

51. What is the role of a query optimizer in an RDBMS? How does it improve query performance?
52. Describe the concept of database sharding and its benefits in scaling a large database.
53. Discuss the concept of database partitioning and its advantages in managing large datasets.
54. What are materialized views in an RDBMS? How do they improve query performance?
55. Explain the concept of transaction isolation levels. What are the different isolation levels, and when are they used?
56. Discuss the concept of database locking. What are shared locks, exclusive locks, and deadlock situations?
57. What is the role of the transaction log in an RDBMS? How does it ensure data consistency and recoverability?
58. Explain the concept of database normalization anomalies (e.g., insertion, update, and deletion anomalies).
59. What are subqueries in SQL? How do they differ from regular queries, and when are they useful?
60. Discuss the concept of database constraints, such as unique constraints, check constraints, and foreign key constraints.
61. Explain the process of database backup and recovery. What are the different backup strategies and their trade-offs?

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